Lead Scoring Case Study Summary:-

The objective of this project was to help X Education improve its lead conversion rate by identifying potential customers who are more likely to convert. The company currently experiences a low conversion rate of around 30%, and the goal was to increase this by prioritizing high-quality leads.

The dataset consisted of approximately 9000 observations with multiple features such as lead source, time spent on the website, total visits, and user activity. The target variable was 'Converted', indicating whether a lead was successfully converted into a customer.

The first step involved data cleaning and preprocessing. Irrelevant columns such as Prospect ID and Lead Number were removed. Missing values were handled appropriately by replacing 'Select' entries with null values and imputing missing data using median and mode techniques. Categorical variables were converted into dummy variables to make the dataset suitable for modeling.

Exploratory Data Analysis (EDA) was performed to identify key patterns. It was observed that leads who spent more time on the website and visited multiple times had a higher likelihood of conversion. Additionally, certain lead sources contributed more effectively to conversions.

A Logistic Regression model was built to predict the probability of conversion. This model was chosen because it provides probabilistic outputs, which were further used to generate a lead score ranging from 0 to 100. The model performance was evaluated using metrics such as accuracy and ROC-AUC score.

To align with the business requirement of achieving a higher conversion rate, threshold tuning was applied. By increasing the threshold, only high-quality leads were selected, thereby improving the efficiency of the sales team.

The results indicate that focusing on high-scoring leads can significantly improve conversion rates. The company can use this model to prioritize leads, optimize marketing strategies, and reduce unnecessary efforts on low-potential customers.

Overall, this project provided valuable insights into customer behavior and demonstrated how data-driven decision-making can enhance business performance.